Ann Oper Res (2019) 283:897–916
https://doi.org/10.1007/s10479-018-2758-6

S.I.: APPLICATIONS OF OR IN DISASTER RELIEF OPERATIONS

# Robust reliable humanitarian relief network design: an integration of shelter and supply facility location

**Mohsen Yahyaei[1] · Ali Bozorgi-Amiri[2]**

Published online: 19 January 2018
© Springer Science+Business Media, LLC, part of Springer Nature 2018

**Abstract** The human societies are threatened by natural disasters. Thus, preparedness and response planning is necessary to eliminate or mitigate their negative effects. Relief network design plays an important role in the efficient response to the affected people. This paper addresses the problem of relief logistics network design under interval uncertainty and the risk of facility disruption. A mixed-integer linear programming model is proposed (1) to consider distribution center (DC) disruption (2) to support the disrupted DC by backup plan (3) to take in to the account both supply and evacuation issues (4) and finally, to mitigate disruption impact by investment. Moreover, robust optimization methodology is applied to hedge against uncertain environments. We conduct computational experiments by using generated instances and a real-world case to perform sensitivity analysis and provide managerial insights. The results show that the total cost of relief network increases by increasing the conservatism level. Moreover, the result show that the total cost of the network can be decrease by reducing the interval of uncertain parameters. As a result, providing more information and better estimation about uncertain parameters can reduce network costs. Disruption probability effect is also investigated and the result indicates that the network tries to establish more reliable facilities as the disruption probability increases. To demonstrate superiority of reliable network described in this paper over the classic network, a Monte Carlo procedure is used to compare two networks and results confirmed superiority of reliable network.

**Keywords** Humanitarian relief logistics · Facility location · Disruption · Robust optimization

## 1 Introduction

Natural disasters have been threatening human societies, and can be costly in terms of human lives and financial losses. Preparedness and response planning is clearly necessary to elimi-

✉ Ali Bozorgi-Amiri
alibozorgi@ut.ac.ir

1 Department of Industrial Engineering, Iran University of Science and Technology, Tehran, Iran

2 School of Industrial Engineering, College of Engineering, University of Tehran, Tehran, Iran

nate or mitigate these loses. One of the critical issues in planning stage is appropriate design of humanitarian relief network. Finding the optimal locations to store and distribute relief items and providing emergency shelters are the key decisions that effect on responsibility of the disaster management system.

Trunick (2005) reported that 80% of relief operations are related to the logistics. Altay and Green provided a list of these logistics operations such as acquiring and stockpiling necessary items, acquiring, stockpiling, and maintaining emergency supplies, constructing central and regional emergency operations centers, evacuating disaster areas, opening shelters and providing mass care (Altay and Green 2006) which most of them are directly related to location of facilities and network design.

The relief network design consists of broad range of model. Huang et al. (2010) developed a p-median problem to response a large scale emergencies. Duhamel et al. (2016) proposed a nonlinear model to locate and distribute of relief items while the affected population has a dynamics nature and a nonlinear behavior.

Barzinpour and Esmaeili (2014) developed a two-echelon relief supply chain. They formulated the problem as a tri-objective model and applied goal-programming approach. The objectives were maximization of covered demands, minimization of facility set up cost and minimization of transportation, holding and shortage cost.

Uncertain environment under disaster conditions are investigated in literature. An pioneer study has been done by Balcik and Beamon (2008). They proposed a scenario based stochastic model to design humanitarian relief chain. Their model considered several coverage levels of demand points and amount of relief items should be stored. The model objective consists of pre-disaster costs such as facility establishment cost and post-disaster costs such as transportation cost of relief items to the affected area.

Döyen et al. (2012) presented a two-stage stochastic model for humanitarian relief logistics problem. The model determines the pre- and post-disaster decision. Location of regional rescue centers and amount of relief items should be stocked (Pre-disaster decisions) are determined in the first stage. In the second stage, model determines location of local rescue centers and transportation of rescue items. They proposed a Lagrangian relaxation method improved by local search to solve the problem in reasonable computational time.

Rath and Gutjahr (2014) considered a warehouse location-routing problem to supply affected area after occurrence of a disaster. They considered several objectives, including minimization of depot set up and transportation cost and maximization of covered demands. To handle multiple objectives, they used adaptive epsilon-constraint algorithm, and they proposed math-heuristic technique to solve large-scale problems. Talebian (2015) considered transportation problem in the aftermath of a disaster. In order to distribute relief resources, they extended open vehicle routing problem. In their model, the affected people should receive the relief items directly or based on a coverage distance. The model aimed to minimize the total transportation cost.

Hong et al. (2015) addressed two models to design humanitarian relief network. In the first model, they proposed a two-stage stochastic model to decide on the size and location of facilities and distribution of relief commodity in a network. In the second model, they extended a chance constraint model that guarantees the demand should be satisfied more than a predetermined probability. Then, they extended the chance constraint model based on the responsiveness and equity.

In addition to the above mentioned approaches, there has been much research regarded robust approaches for humanitarian network design. The aim of robust optimization approaches are to provide an optimal solution by regarding a specified level of protection against infeasibility. In this context, Najafi et al. (2013) presented a robust multi-objective model for humanitarian relief operations in earthquake response phase. The relief operations include servicing injured people and transportation of relief commodities. Network parameters such as demands and supplies are subjected to interval uncertainty. Their model encompassed multiple objectives, including minimization of total unserved injured people, unsatisfied demands and total utilized vehicles.

Zokaee et al. (2016) proposed a robust model for humanitarian relief logistics. The model aimed to determine the set of facilities to be established while minimizing the total cost of open facilities, transportation and shortage cost. They considered the interval uncertainty in parameters such as demands and supplies.

On the top of parameters uncertainty, part of network may be disrupted by a disaster. With regard to this issue, a multi-objective model is developed by Bozorgi et al. (2013) to design of humanitarian relief logistics in both pre- and post-disaster stages. Demand and supply are regarded as uncertain parameters. They proposed a robust two-stage stochastic model that tried to minimize expected value and the variance of the total cost over all scenarios.

Ahmadi et al. (2015) developed a two-stage stochastic location-routing problem in the pre- and post-phases of an earthquake occurrence. In the first stage (strategic level), the model determines location of main distribution centers and aggregate demand points. In the second stage model (operational level), once the main distribution centers and aggregate demand points are located, the location of local distribution center and vehicle routing have to be determined. In this stage, vehicle routing is considered under various destruction scenarios. They applied a variable neighborhood search algorithm to solve large instances. Salman and Yücel (2015) proposed a scenario based model to locate emergency facilities. In their model, scenarios were generated by spatial correlation approach named as distance-based dependency. In each scenario, a number of links could fail and model tries to maximize expected covered demand points. They proposed a Tabu search algorithm to solve the case study motivated from Istanbul earthquake preparedness.

Tofighi et al. (2016) considered a two-stage stochastic humanitarian relief network problem under mixed uncertainty. Decision on location of facilities and amount of stored goods are taken in the first stage. In the second stage, a scenario based multi objective problem is developed for minimization of total distribution times, maximum weighted travel time and cost of unused items or shortage. They also considered destruction of facilities and transportation routes in each scenario.

Bozorgi-Amiri and khorsi (2016) proposed a scenario based multi-objective model to design of humanitarian relief logistics. The proposed model determined location of facilities, allocation of suppliers and transportation routes as strategic, tactical and operational decisions, respectively. In their model, they considered roads and facilities disruption scenarios.

Strategies of supplying the relief items are considered by researchers. For example reservation of supplies in pre-disaster phase (Yang et al. 2016), regarding perishability of stored stock (Diatha and Sundar 2015), the sustainable procurement (Kaur and Singh 2016) are investigated.

Evacuation planning is the other major operations in relief logistics which received much attention in the literature. Kilci et al. (2015) developed a mixed integer model to locate temporary shelters and controlling utilization of shelters after occurrence of an earthquake. Li et al. (2012) proposed a stochastic bi-level model to locate shelters and transportation of evacuees to shelters. Sheu and Pan (2014) developed a three-stage integrated model to design shelter, medical and distribution sub-networks in the centralized emergency supply network. Zhen et al. (2015) developed an integrated model to locate new supply facility and shelters while some facilities are existed in advanced. Jin et al. (2015) considered transportation of victims with different type of injuries. Their propose model tried to maximize number of survivors. From reviewed research, it can be concluded that:

(1) A limited number of researches considered both supply and evacuations issues in an integrated model.
(2) There is no study that considered both robust approach and reliability of supply infrastructures in the context of humanitarian relief network design.

This study aims to fill these research gap by developing a new mathematical model to design the relief logistics network in presence of disruption and interval demand uncertainty. Thus, the contributions of this paper are:

(1) Consideration of implicit formulation of disruption: All the cited papers considered scenario-based disruption in the network. As it is clear, computational time in the scenario-based formulation is strongly dependent on the number of scenarios, and optimal solution may not be obtained when the number of scenarios increases.
(2) Robust Integrated evacuation and supply facility location: After occurrence of disaster, it is necessary to evacuate people into proper place, and such facility should be considered in advance and in pre-disaster phase. Therefore, an integrated supply and evacuation model is developed in this paper. Moreover, we considered standard evacuation distance constraint in the evacuation operations, which is equivalent to the standard relief time constraints proposed in Ahmadi et al. (2015).
(3) Considering a robustness and reliability aspects for humanitarian network design
(4) Consider of supportive distribution center and backup and indirect distribution: After disruption of a distribution center (DC), backup facility is considered to support them. We call them supportive distribution center (SDC), and they help the disrupted DC instead of direct servicing of its allocated shelters. The reason of indirect distribution is that the personnel of SDC may be unfamiliar to affected area supported by the disrupted DC, and it is challenging and difficult task for SDC personnel to distribute relief items fairly and correctly (See Pan American (2001) for more details about direct and indirect distribution). Similar strategy is applied in the commercial logistics which is called as goods sharing strategy (Azad et al. 2013; Hatefi et al. 2015). To the best of known, this strategy is not considered in humanitarian relief logistics.

Moreover, in the following, we highlight the other modeling features of the relief logistics network:

(1) Budget based disruption mitigation: Impact of damage on the network can be mitigated by using safer structure. In this paper, more investment on facility leads to better disaster-resistant facility. The investment strategy applied in this paper is motivated from soft hardening strategy proposed in Azad et al. (2013).
(2) Robust network design: The network parameters cannot be exactly estimated after occurrence of disaster and the parameters uncertainty cannot be neglected. To deal with the uncertainty, the robust optimization is applied in this paper similar to Najafi et al. (2013) and Zokaee et al. (2016).

The rest of this paper is organized as follows. In the next section, the problem is formulated. Then, the robust optimization is reviewed in Sect. 3. We further propose robust counterpart of the proposed model in Sect. 3. Computational results and analysis are provided in Sect. 4. In this section, an application of proposed relief logistics is reported. Finally, the concluding remarks are presented in Sect. 5.

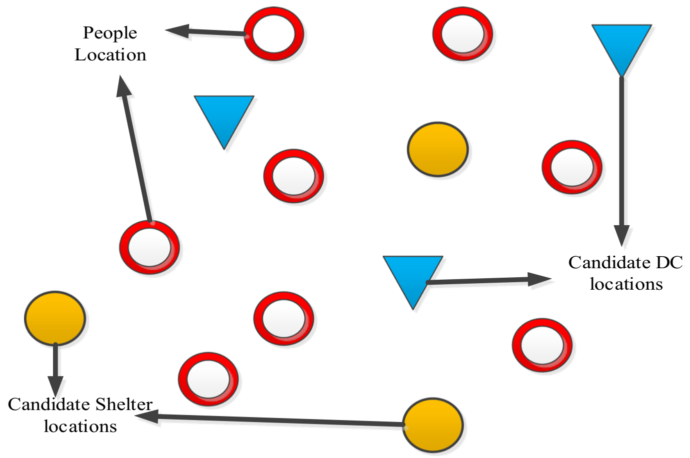


**Fig. 1** Available network in pre-disaster phase to locate facilities

## 2 Problem description

This paper deals with a robust reliable humanitarian relief network design problem. There are three types of facilities which can be located in the network:

(1) SDC: The safe infrastructure which is designed to supply relief items for the affected people in the allocated shelters. Moreover, they are considered as a heliport which can deploy relief resources of unaffected near cities (particularly in short-term). In this paper, we considered them as incapacitated facilities.
(2) Unreliable distribution center (UDC): The relief items could be stocked at there in pre-disaster phase. They may be disrupted based on pre-specified probability. However, their vulnerability could be controlled by level of investment (can be varied from fully to partially disrupted one in case of disruption). They are capacitated facilities, and SDCs are considered as their backups.
(3) Shelter: the capacitated temporary facilities to settle evacuees. An SDC or UDC can supply this facility.

An available network is demonstrated in the Fig. 1 before establishing facilities. This paper aims to design a robust reliable relief network to hedge against uncertainty of parameters and disruption of UDCs. After the disaster, disruption may occur in a UDC, and an SDC supports it instead of allocated shelter to disrupted one (indirect shipment). In the Fig. 2, a designed network is depicted.

The following notations are used to formulate the model:

### 2.1 Indices and set

| | |
|---|---|
| $N$ | Set of demand node indexed by $i.j \in N$ |
| $A$ | Set of candidate shelters indexed by $j \in J$ |
| $ML$ | Set of candidate DCs indexed by $k$ and $k' \in K$ |
| $MF$ | Set of investment levels for opening a UDC indexed by $n \in N$ |

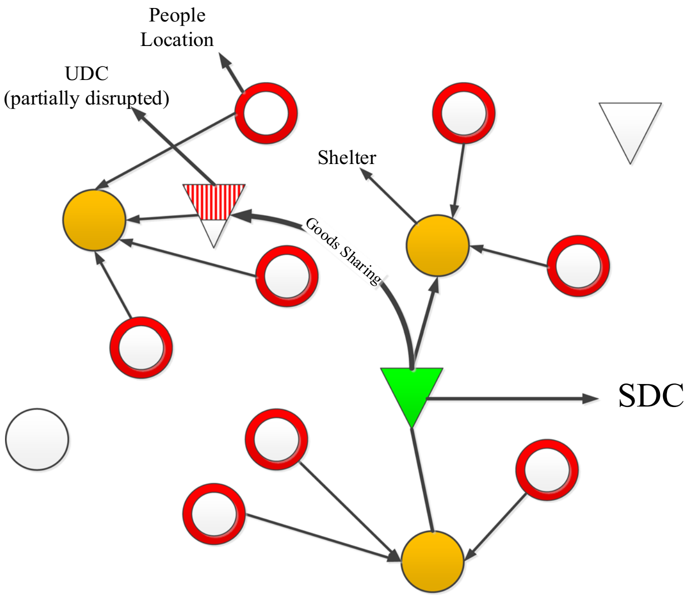


**Fig. 2** A designed network in post-disaster phase

## 2.2 Parameters

| | |
|---|---|
| $\tau$ | Standard evacuation distance |
| $\delta_{ij}$ | Distance between people in location $i$ to shelter at $j$ |
| $c_{ij}$ | Transportation cost from people in location $i$ to shelter at $j$ |
| $c'_{jk}$ | Transportation cost for a unit of relief item from SDC/UDC at $k$ to Shelter at $j$ |
| $c''_{kk'}$ | Transportation cost for a unit of relief item from SDC at $k$ to UDC at $k'$ |
| $a_i$ | Number of affected people in location $i$ |
| $b_j$ | Capacity of Shelter at $j$ |
| $FS_j$ | Fixed cost of opening shelter at $j$ |
| $Fsdc_k$ | Fixed cost of opening SDC at $j$ |
| $Fudc_{kn}$ | Fixed cost of opening UDC at $j$ with investment level $n$ |
| $q_k$ | Disruption probability in UDC at $k$ |
| $e_k$ | Capacity of UDC at $k$ |
| $rc_{kn}$ | Percentage of total capacity of disrupted UDC at $j$ opened with investment level $n$ |
| $M$ | A large enough positive constant |
| $ff$ | Conversion factors person to food |

## 2.3 Decision variables

| | |
|---|---|
| $x_{ij}$ | 1 if people in location $i$ allocate to shelter at $j$; 0,otherwise |
| $yr_{jk}$ | Fraction of relief items demand of shelter at $j$ supported by SDC at $k$ |
| $yu_{jk}$ | Fraction of relief items demand of shelter at $j$ supported by USDC at $k$ |
| $u_j$ | 1 if shelter is opened at $j$; 0, otherwise |

| | |
|---|---|
| $v_k^r$ | 1 if SDC is opened at $k$;0,otherwise |
| $v_{kn}^u$ | 1 if UDC is opened at $k$ with investment level $n$;0, otherwise |
| $t_{ij}$ | Amount of shipped relief items from SDC at $k$ to UDC at $j$ at disruption situation |

## 2.4 Problem formulation

We present the model as follows (Model 1):

$$\min \sum_i \sum_j c_{ij} x_{ij} a_i + \sum_j \sum_k c'_{jk} y_{jk}^r b_j + \sum_j \sum_k c'_{jk} b_j y_{jk}^u + \sum_j Fs_j u_j$$

$$+ \sum_k Fsdc_k v_k^r + \sum_k \sum_n Fudc_{kn} v_{kn}^u + \sum_k \sum_{k'} \sum_n q_{k'} t_{kk'} c''_{kk'} \tag{1}$$

$$\sum_i x_{ij} \le M \times u_j \quad \forall j \tag{2}$$

$$\sum_i x_{ij} = 1 \quad \forall j \tag{3}$$

$$\delta_{ij} x_{ij} \le \tau \quad \forall i, j \tag{4}$$

$$\sum_j y_{jk}^r \le M \times v_k^r \quad \forall k \tag{5}$$

$$\sum_i a_i x_{ij} \le \sum_k b_j (y_{jk}^r + y_{jk}^u) \quad \forall j \tag{6}$$

$$\sum_j b_j y_{jk}^u ff \le e_k \sum_n v_{kn}^u \quad \forall k \tag{7}$$

$$\sum_n v_{kn}^u + v_k^r \le 1 \quad \forall k \tag{8}$$

$$\sum_k v_k^r \ge 1 \tag{9}$$

$$t_{kk'} \le \sum_j b_j y_{jk'}^u \quad \forall k, k' \tag{10}$$

$$t_{kk'} \le v_k^r \quad \forall k, k' \tag{11}$$

$$\sum_k t_{kk'} + e_{k'} (\sum_n rc_{k'n} v_{k'n}^u) \ge \sum_j b_j y_{jk'}^u \quad \forall k' \tag{12}$$

$$t_{kk'}, y_{jk}^u, y_{jk}^r \ge 0 \quad \forall k, k', j \tag{13}$$

$$x_{ij}, v_{kn}^u, v_k^r, u_j \in \{0, 1\} \quad \forall i, j, k, n \tag{14}$$

The objective function (1) minimizes total cost of evacuees transportation to shelter (the first term), transportation of relief items to shelter from SDC/UDCs (the second/third terms), shelters, SDCs and UDCs opening cost (the fourth, fifth and sixth terms, respectively) and expected transportation cost from the backup SDC to disrupted UDC cost (the seventh term).

Constraints (2) assure that evacuees in each location can only be assigned to opened shelters. Constraints (3) restrict each demand point to be allocated to one shelter. Constraints (4) assure that evacuees can be allocated to shelters within less than standard evacuation distance. Constraints (5) indicate that a shelter can be served by an opened SDC. Constraints (6) and (7) show capacity constraints, constraints (6) ensure that number of evacuees allocated to the shelter at $j$ cannot exceed the shelter capacity while constraints (7) guarantee total volume of demanded items by shelters in the UDC at $k$ cannot exceed the capacity of the UDS. Constraints (8) ensure that supply facilities (SDC or UDC) cannot be established at the same location. Constraint (9) shows that at least an SDC should be opened in the network. Constraints (10) show that amount of goods that SDCs send to support a disrupted UDC cannot exceed from capacity of the UDS. Constraints (11) guarantee that supportive relief items can only originate from an SDC. Constraints (12) show relation of goods sharing and soft hardening strategies. These constraints represent that amount of required supportive relief items and remained relief items in a disrupted UDC should satisfy allocated demands to the UDC. As it is clear, the higher rate of $rc$ leads to be in need of fewer supportive relief items. Constraints (13) impose positivity restriction and constraints (14) enforce the binary restriction on the corresponding decision variables.

## 3 Robust optimization

Most real-world optimization problems involve uncertainties. Stochastic programming is a well-known approach to handle uncertainty. However, there may be constraint violation in the stochastic programming (Ben-Tal and Nemirovski 1999). To handle this problem, the robust stochastic programming using the concept named as soft constraint can be applied (Mulvey et al. 1995). This means that model can consider constraint violation by a penalty. Nevertheless, in vital systems like relief operations, it is needed to consider hard constraints without violation. Moreover, in some cases, there may be no distribution function to characterize uncertain parameters of a system, and they can only be described by an interval without any probability distribution information. In the design of relief logistic network, it is difficult to estimate parameters of network such as the number of affected people by a probability distribution function. In contrast to stochastic programming, robust optimization approach can handle both hard constraints and interval uncertainty. In robust optimization, an interval of uncertainty is considered for parameters, and the aim is to find a solution to be feasible for all realization of uncertain parameters. In this paper, we implement robust optimization approach proposed by Bertsimas and Sim (2004). A brief introduction of this approach is provided below.

Consider the following problem (P1):

$$\begin{aligned} &\max Cx \\ &Subject\ to: \sum_{j\in J_i} \tilde{a}_{ij}x_j + \sum_{j\in N\backslash\{J_i\}} a_{ij}x_j \le b_i \quad \forall i, \tilde{a}_{ij} \in J_i \\ &0 \le x \end{aligned} \tag{15}$$

where $J_i$ is the set of uncertain parameters of the $i$th row and $\tilde{a}_{ij} \quad j \in J_i$ is the uncertain parameter that takes value according to a symmetric distribution in the interval $[a_{ij} - \hat{a}_{ij}, a_{ij} + \hat{a}_{ij}]$ where $a_{ij}$ is nominal value (mean of the interval), and $\hat{a}_{ij}$ is a constant deviation of $\tilde{a}_{ij}$. In this approach, parameter $\Gamma_i$ is considered for every $i$ which is named as budget of uncertainty and it controls the conservatism level of solution. $\Gamma_i$ can take a value in the interval $[0, |J_i|]$. In the present study, we consider integer value of $\Gamma_i$. Budget of uncertainty guarantees feasibility of solution up to simultaneous change of $\Gamma_i$ uncertain parameters.

Based on the mentioned principles, P1 can be formulated as follows (P2):

$$\begin{aligned} &\max Cx \\ &Subject\ to: \max_{\{S_i|S_i\in J_i, |S_i|=\Gamma_i\}} \{\sum_{j\in S_i} \hat{a}_{ij}x_j\} + \sum_{j\in N} a_{ij}x_j \le b_i \quad \forall i \\ &0 \le x \end{aligned} \tag{16}$$

It is clear when $\Gamma_i = 0$, P2 is equivalent to P1 with nominal value. P2 is a nonlinear model and it can be transformed to linear one as follows (P3) (Bertsimas and Sim 2004).

$$\begin{aligned} &\max Cx \\ &Subject\ to: z_i\Gamma_i + \sum_{j\in J_i} p_{ij} + \sum_{j\in N} a_{ij}x_j \le b_i \quad \forall i \\ &z_i + p_{ij} \ge \hat{a}_{ij}x_j \quad \forall i, j \in J_i \\ &0 \le x, z, p \end{aligned} \tag{17}$$

### 3.1 The robust formulation of model 1

We consider the number of affected people as interval uncertain parameters with symmetric distribution ($\tilde{a}_i \in [a_i - \hat{a}_i, a_i + \hat{a}_i]$). Accordingly, Eq. (1) and (6) in *Model 1* change as follows:

$$\begin{aligned} \min \sum_i \sum_j c_{ij}x_{ij}\tilde{a}_i + \sum_j \sum_k c'_{jk}y^r_{jk}b_j + \sum_j \sum_k c'_{jk}b_j y^u_{jk} + \sum_j Fs_j u_j \\ + \sum_k Fsdc_k v^r_k + \sum_k \sum_n Fudc_{kn}v^u_{kn} + \sum_k \sum_{k'} \sum_n q_{k'}t_{kk'}c''_{kk'} \end{aligned} \tag{18}$$

$$\sum_i \tilde{a}_i x_{ij} \le \sum_k b_j(y^r_{jk} + y^u_{jk}) \quad \forall j \tag{19}$$

The robust counterpart of Eqs. (18) and (19) are Eqs. (20)–(24).

$$\begin{aligned} \min \sum_i \sum_j C_{ij}x_{ij}a_i + \Gamma_0 z_0 + \sum_i p_{0i} \\ \sum_j \sum_k C'_{jk}y^r_{jk}b_j + \sum_j \sum_k C'_{jk}b_j y^u_{jk} + \sum_j Fs_j U_j \\ + \sum_k Frdc_k V^r_k + \sum_k \sum_n Fudc_{kn}V^u_{kn} + \sum_k \sum_{k'} \sum_n q_{k'}T_{kk'}C'_{kk'}V^u_{k'n} \end{aligned} \tag{20}$$

$$\sum_i a_i x_{ij} + \Gamma_j z_j + \sum_i p_{ij} \le bb_j U_j \quad \forall j \tag{21}$$

$$z_0 + p_{0i} \ge \hat{a}_i \sum_j C_{ij}x_{ij} \quad \forall i \tag{22}$$

$$z_j + p_{ij} \ge \hat{a}_i x_{ij} \quad \forall i, j \tag{23}$$

$$z_j, p_{ij} \ge 0 \tag{24}$$

## 4 Computational experiments

In this section, we conduct numerical experiments to understand the impact of disruption and uncertainty on the model. The computational results were obtained using GAMS/CPLEX on a laptop computer with an INTEL Core i5 CPU with 2.5GHz clock speed and 4 GB of RAM. This section divided into two subsections. In the first subsection, we analyze behavior of the proposed model. Then, in the second subsection, its practical application is reported.

**Table 1** Source of parameters generation used for the network with 25 nodes

| Parameters | Corresponding value |
|---|---|
| $a_i$ (nominal value) | ~U (100, 1000) |
| Land cost | ~U (1000, 10,000) |
| $\tau$ | 50 |
| $rc_{kn}$ | |
| 1 | 0 |
| 2 | 0.3 |
| 3 | 0.5 |
| $ff$ | 1.5 |
| $FS_j$ | $Landcost \times 7.5$ |
| $Fudc_{kn}$ | |
| 1 | $Landcost \times 15$ |
| 2 | $Landcost \times 17.5$ |
| 3 | $Landcost \times 20$ |
| $Fsdc_k$ | $Landcost \times 25$ |
| $e_k$ | 8500 |
| $b_j$ | 2600 |
| $c_{ij}$ | $\delta_{ij}$ |
| $c'_{jk}$ | $\delta_{ij} \times 0.75$ |
| $c''_{kk'}$ | $\delta_{ij} \times 0.75$ |

### 4.1 Model analysis

To analyze proposed model, a network with 25 nodes and random parameters is designed according to Table 1. In this example, all 25 nodes are considered as a candidate to open shelters and DCs. These 25 nodes spread over an area of 100 by 100 units. Land cost is considered for each point and the opening cost of each type of facilities is multiplied by the land cost. Three levels for UDCs are considered and by increasing investment level, the opening cost of facility increases and $\delta_{ij}$ equals Euclidean distance between nodes $i$ and $j$.

#### *4.1.1 Impact of uncertainty on network performance*

To analyze impact of uncertainty, disruption probability is randomly generated from the interval [0.05, 0.2]. To study the impact of uncertainty on the network performance, $\hat{a}_i$ is considered as 5, 20 and 35 percentage of $a_i$. $\Gamma$ is also considered at various levels. The experiment results are reported in Table 2. As the results show, by increasing $\Gamma$ and $\hat{a}_i$, the network cost goes up. The results also indicate that computational time in instances with higher value of $\Gamma (\Gamma = 10, 13, 16, 25)$ is more than instances with low value of $\Gamma (\Gamma = 1, 4, 7)$ and the nominal case ($\Gamma = 0$). It is clear that gathering more information about uncertain parameter can reduce the interval of uncertain parameters in practical cases and subsequently, it can reduce the total network cost.

Moreover, Fig. 3 shows that the network cost is very sensitive to small change of $\Gamma$ in the lower value. As the figure shows, when the $\Gamma$ is set zero, network cost under different values of $\hat{a}_i$ are equal to the nominal value.

**Table 2** Impact of uncertainty on the model the network performance and computational time

| Deviation($\hat{a}_i$) | 0.05 | | 0.2 | | 0.35 | |
|---|---|---|---|---|---|---|
| | OFV | TIME | OFV | TIME | OFV | TIME |
| *Budget of uncertainty* ($\Gamma$) | | | | | | |
| 0 | 696809.2 | 11.932 | 69609.2 | 11.969 | 696809.2 | 12.015 |
| 1 | 769646.3 | 12.297 | 775126.7 | 11.846 | 780643.8 | 12.997 |
| 4 | 955159.2 | 34.025 | 972254.8 | 31.79 | 989029.2 | 44.144 |
| 7 | 1016078 | 43.397 | 1041278 | 43.719 | 1065970 | 47.091 |
| 10 | 1032777 | 48.373 | 1062849 | 48.511 | 1091348 | 49.526 |
| 13 | 1047274 | 65.828 | 1073716 | 41.727 | 1096037 | 43.774 |
| 16 | 1047481 | 57.983 | 1074079 | 51.568 | 1097448 | 34.881 |
| 25 | 1047481 | 59.076 | 1074079 | 43.504 | 1097448 | 30.127 |

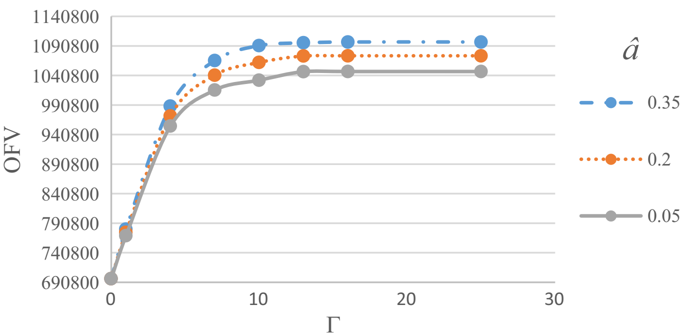


**Fig. 3** Relationship between the network cost and $\Gamma$

### *4.1.2 Impact of disruption probability on the network performance*

To analyze impact of disruption probability on network performance, different disruption probabilities ($q_k$ = 0,0.025,0.05,0.075,0.09,0.1,0.2,0.3), $\Gamma = 5$ and $\hat{a}_i = 0.05 \times a_i$ are considered. The depicted results in Fig. 4 show that the expected network cost increases by increasing the disruption probability. Moreover, in Fig. 4, the "Relative Cost" series represents the percentage difference between the expected network cost and based cost (network cost with $q_k = 0$). Figure 4 shows that the network is more sensitive to the changes in small value of disruption probability.

### *4.1.3 Impact of disruption probability and budget of uncertainty on the network configuration*

To study the impact of both disruption probability and $\Gamma$ on the network configuration (the type and number of facilities), different levels of $\Gamma$ and disruption probability are considered according to Table 3. The obtained results show that by increasing $\Gamma$, the network attempts to open more facilities in all types of applications to tackle with uncertainty. The results show that disruption probability only effects on types of DCs. The model tries to investment on

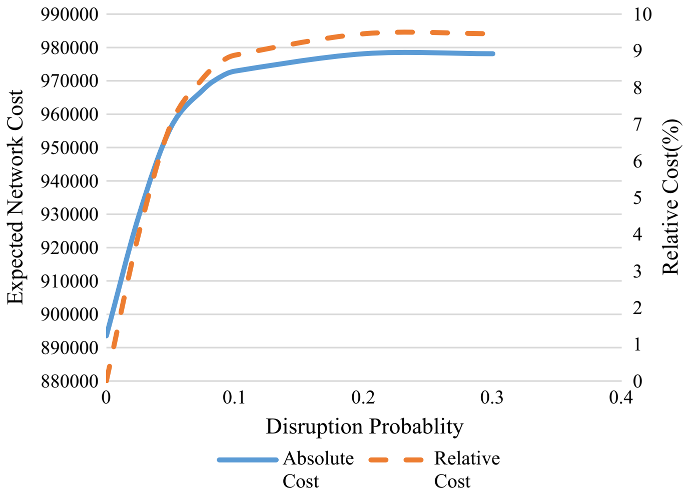


**Fig. 4** Relationship between the network cost and disruption probability

more reliable facilities as disruption probability increases. Figure 5 shows the trend of model towards more reliable network (with $\Gamma = 5$). This figure illustrates that when the disruption probability is 0.025, five UDCs at level 1 and a SDC are established in the optimal design. By increasing disruption probability, the network tries to establish more reliable infrastructures. For example, when the disruption probability is 0.5, two UDCs at level 3 and three SDCs are established.

By increasing both $\Gamma$ and disruption probability, the model design more reliable network with more number of facilities.

#### *4.1.4 Benefit of considering disruption on network design*

Our proposed model is developed based on minimizing the expected cost of the network and the model make decision based on all realization. The facilities are established in the network before occurrence of the disaster. When a disaster occurred, a disruption may be realized. However, it is clear that the designed network may not be the optimal decision for a particular realization after disaster occurs.

In this section, Reliable network is compared with classical network (without consideration of disruption) by generating different future realizations, and the benefit of reliable design is investigated. To do so, classical network is considered similar to Model 1 without supportive shipment variable ($t_{ij}$) and Eqs. (10)–(12). Then, both networks solve and the location of facilities is recorded for both networks (the pre-disaster decisions), and we consider them as fixed variable. To evaluate expected network cost, all possible scenarios should be considered. There are $2^{25}$ possible scenarios can occur in the network. To tackle numerous scenarios, Monte Carlo procedure is implemented similar to Azad et al. (2013). By this procedure, 1000 random disruption scenarios are generated (post-disaster conditions) to compare two networks efficiency. If disruption occurs on a UDC in a scenario, the lost capacity will be provided by the nearest SDC. Expected costs of these two networks are reported in Table 4. The results show that when disruption probability equals to zero, two networks are the same (the first rows of reported results). However, when disruption probability increases, the reliable network considerably works better than the other one.

**Table 3** Computational results of disruption probability and budget of uncertainty impact on the network configuration

| | $\Gamma = 0$ | | | | | $\Gamma = 5$ | | | | | $\Gamma = 10$ | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Shelter | Unreliable DC level | | | SDC | Shelter | Unreliable DC level | | | SDC | Shelter | Unreliable DC level | | | SDC |
| Disruption probablity | | 1 | 2 | 3 | | | 1 | 2 | 3 | | | 1 | 2 | 3 | |
| 0 | 6 | 3 | 0 | 0 | 1 | 9 | 5 | 0 | 0 | 1 | 10 | 5 | 0 | 0 | 1 |
| 0.025 | 6 | 3 | 0 | 0 | 1 | 9 | 5 | 0 | 0 | 1 | 10 | 5 | 0 | 0 | 1 |
| 0.05 | 6 | 2 | 1 | 0 | 1 | 9 | 2 | 1 | 0 | 2 | 10 | 4 | 1 | 0 | 1 |
| 0.06 | 6 | 2 | 2 | 0 | 1 | 9 | 1 | 2 | 0 | 2 | 10 | 3 | 2 | 0 | 1 |
| 0.075 | 6 | 2 | 2 | 0 | 1 | 9 | 1 | 2 | 0 | 2 | 10 | 1 | 2 | 0 | 2 |
| 0.09 | 6 | 0 | 1 | 2 | 1 | 9 | 1 | 0 | 1 | 3 | 10 | 1 | 0 | 1 | 3 |
| 0.1 | 6 | 0 | 0 | 3 | 1 | 9 | 1 | 0 | 1 | 3 | 10 | 1 | 0 | 1 | 3 |
| 0.2 | 6 | 0 | 0 | 3 | 1 | 9 | 0 | 0 | 2 | 3 | 10 | 0 | 0 | 1 | 4 |
| 0.3 | 6 | 0 | 0 | 2 | 2 | 9 | 0 | 0 | 2 | 3 | 10 | 0 | 0 | 1 | 4 |
| 0.5 | 6 | 0 | 0 | 2 | 2 | 9 | 0 | 0 | 2 | 3 | 10 | 0 | 0 | 1 | 4 |
| 0.8 | 6 | 0 | 0 | 2 | 2 | 9 | 0 | 0 | 2 | 3 | 10 | 0 | 0 | 1 | 4 |

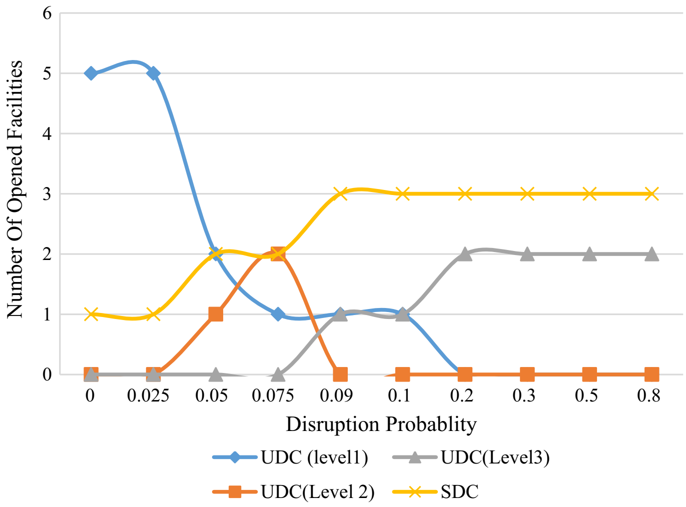


**Fig. 5** Impact of disruption probability on the network supply facilities ($\Gamma = 5$)

**Table 4** Comparison of reliable and classic network

| Disruption probablity | $\Gamma = 0$ | | $\Gamma = 5$ | | $\Gamma = 10$ | |
|---|---|---|---|---|---|---|
| | Classic system | Reliable system | Classic system | Reliable system | Classic system | Reliable system |
| 0 | 640516.352 | 640516.352 | 939420.559 | 939420.559 | 957086.022 | 957086.022 |
| 0.05 | 697687.692 | 679349.541 | 1012992.396 | 977507.763 | 1061769.044 | 1026561.857 |
| 0.075 | 719109.629 | 691779.108 | 1052731.22 | 984425.579 | 1129883.99 | 1056345.986 |
| 0.1 | 747362.19 | 698440.728 | 1110011.978 | 992861.835 | 1158406.426 | 1063988.804 |
| 0.15 | 807802.166 | 699384.344 | 1197986.011 | 997345.182 | 1272829.401 | 1072028.902 |
| 0.2 | 855493.288 | 702485.321 | 1292023.434 | 999256.78 | 1357685.957 | 1075175.162 |
| 0.3 | 971599.697 | 704541.352 | 1496421.439 | 999067.132 | 1573740.804 | 1075175.162 |
| 0.4 | 1093773.545 | 704541.352 | 1690374.969 | 999067.132 | 1813475.17 | 1075175.162 |

Now, to compare two networks in different future situations, the relative percentage deviation (RPD) is used as a common performance measure as follows:

$$RPD = \frac{OFV_{Classic\,network} - OFV_{Reliabale\,network}}{OFV_{Classic\,network}} \times 100 \tag{25}$$

The obtained results show that the reliable network consistently provides better solutions and it significantly outperforms in terms of RPD in the high rate of disruption probability (Fig. 6).

## 4.2 Application

In this subsection, the proposed model is implemented in the 1st district of Tehran as a case study proposed by Barzinpour and Esmaeili (2014) (Fig. 7). The earthquake is a potential disasters that threats Tehran. Vulnerability analysis under earthquake of magnitude of 7.2 on the Richter scale by the fault located in north of Tehran is depicted in Fig. 8.

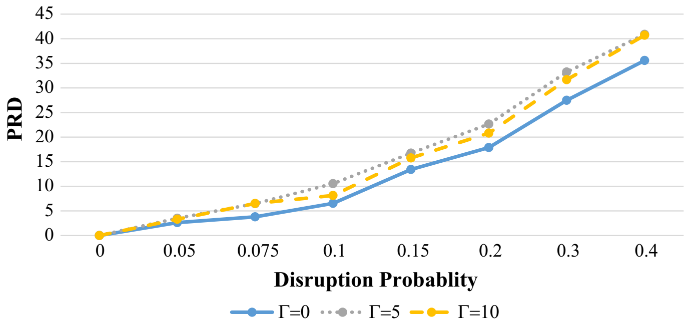


**Fig. 6** Comparison of reliable and classic networks

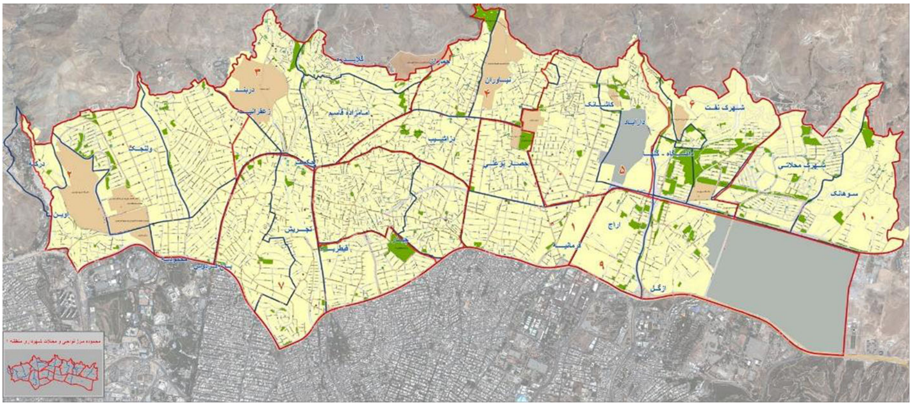

**Fig. 7** Map of the case study

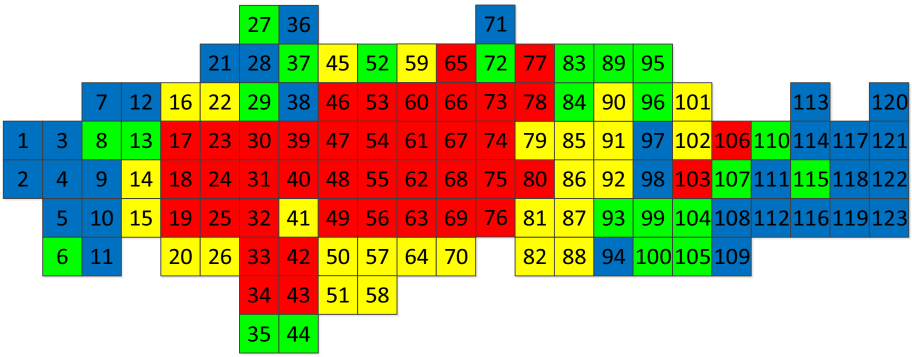


**Fig. 8** The earthquake damage estimation of case study (Barzinpour and Esmaeili 2014)

In this case study, the district is divided into 123 equal square blocks, and these blocks are illustrated by four colors (Fig. 8). The block colors on Fig. 8 show the earthquake damage estimation. The order of each color in terms of vulnerability is red (the most vulnerable), yellow, green and blue (the safest block). Most of the required parameters are provided in this case study. However, the other parameters are estimated or assumed as follows:

**Table 5** The disruption probability of each block based on the associated color

| Block color | Disruption probability |
|---|---|
| Red | 0.20 |
| Yellow | 0.15 |
| Green | 0.10 |
| Blue | 0.05 |

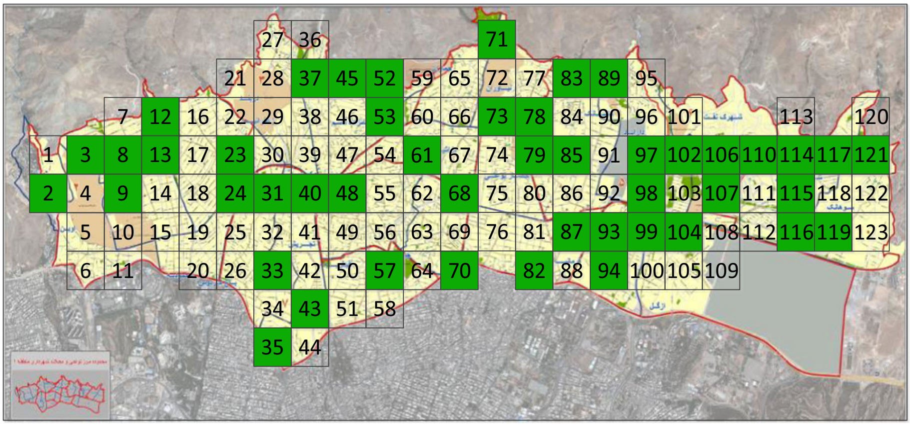


**Fig. 9** Candidate locations for opening shelters (the green blocks). (Color figure online)

(1) The block disruption probability is corresponded to the estimated damage of block and reported in Table 5.

(2) The red blocks are neglected to be considered as candidate location of DC. Similar to Saadatseresht et al. (2009), the parks and open areas are considered as location of candidate shelters, because they can provide basic living requirements such as water, toilet, electricity, enough space for evacuees and they are not threatened by earthquake. The Shelter candidate locations are depicted in Fig. 9.

(3) We assume that each UDC can store 30,000 relief items and capacity of shelters is considered as a factor of the amount of area they can provide.

(4) Two investment levels for UDCs are considered. The UDC opening cost at level 1 and 2 are assumed to be 3 and 4 times the land cost, respectively. The SDC opening cost is considered to be 15 times the land cost. Shelter opening cost is evaluated based on the space they provide and calculate as follows:

$$Fs_i = \left(1 + 2 \times \frac{space_i - Minspace}{Maxspace - Minspace}\right) \times landcost_i \tag{26}$$

where $space_i$, $Minspace$ and $Maxspace$ are amount of space available in candidate shelters location at $i$, minimum and maximum open area existed in the district, respectively.

(5) All transportation cost considered 0.01 unit per meter for all cases and Standard evacuation distance is considered 3000 meters.

The results are reported in Table 6 under different $\Gamma$ values. Similar to the results obtained in the pervious section, by increasing $\Gamma$, the network cost and number of opened facilities increase. Location of opened DCs and shelters are illustrated in the Figs. 10 and 11.

**Table 6** Obtained results of case study under different Γ values

| Γ | Network cost | Shelter location | 1 level unreliable DC | 2 level unreliable DC | Reliable DC |
|---|---|---|---|---|---|
| 1 | 5550063.533 | 3 8 9 12 13 33 37 45 52 53 57 71 79 87 89 93 98 99 102 104 107 114 115 116 119 121 | 8 9 12 13 98 107 114 116 109 | 37 45 52 71 79 89 93 99 104 115 117 121 | 26 57 102 |
| 5 | 7659895.138 | 3 24 53 79 102 117 8 31 57 87 104 119 9 33 61 89 107 121 12 37 68 93 114 13 45 70 98 115 23 52 71 99 116 | 8 9 12 13 70 98 107 114 116 117 119 | 3 37 45 52 79 89 93 99 104 115 | 26 57 71 102 121 |
| 10 | 9365898.805 | 2 8 9 12 13 23 24 31 33 35 37 40 42 45 48 52 53 57 61 68 70 71 73 79 82 85 87 89 93 97 98 99 102 104 107 114 115 116 117 119 121 | 8 9 12 13 35 70 85 87 93 97 98 107 114 116 117 119 | 3 37 41 45 52 79 89 99 104 115 | 26 57 71 102 121 |
| 15 | 1.09E+07 | 2 8 9 12 13 23 24 31 33 35 37 40 42 43 45 48 52 53 57 61 68 70 71 73 78 79 82 85 87 89 93 94 97 98 99 102 104 107 114 115 116 117 119 121 | 8 9 12 13 35 70 85 97 98 107 114 116 117 119 | 3 37 41 45 52 79 89 93 99 104 115 | 26 57 71 102 121 |
| 25 | 1.46E+07 | 2 3 8 9 12 13 23 24 31 33 35 37 40 42 43 45 48 52 53 57 61 68 70 71 73 78 79 82 83 85 87 89 93 94 97 98 99 102 104 107 110 114 115 116 117 119 121 | 8 9 12 13 35 52 70 85 87 97 98 107 110 114 116 117 119 | 3 41 79 89 93 99 104 115 | 26 37 57 71 102 121 |

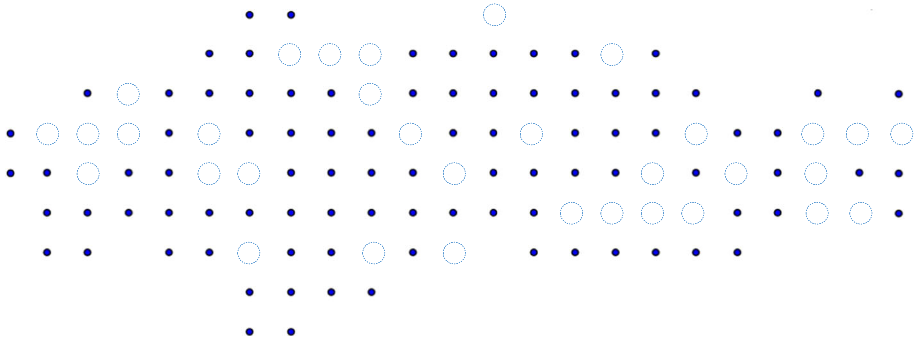

**Fig. 10** Opened shelters under $\Gamma = 5$

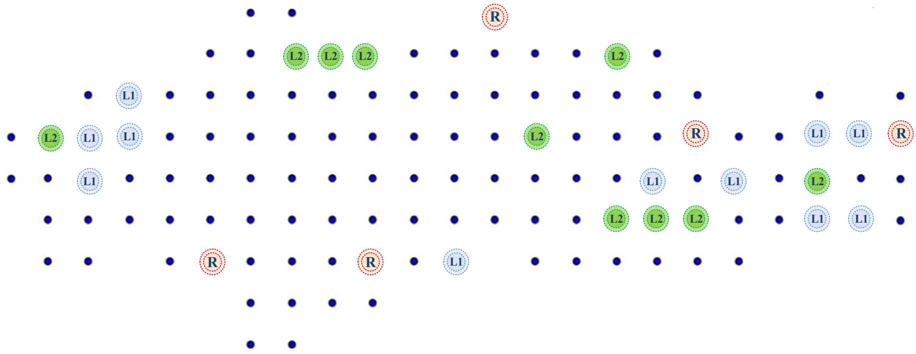


**Fig. 11** Opened UDCs and SDCs under $\Gamma = 5$

## 5 Managerial implications

The proposed model provides a number of managerial implications to support relief logistics designers. Firstly, the proposed model considered robustness and reliability which are major issues in the context of relief network design. In the operational level and distribution of relief item phase, proposing an indirect distribution may be highly effective especially for designing relief network of Mega cities.

Secondly, several aspects of the model such as the impact of disruption and parameters uncertainty were studied to help and guide network designer for establishing a proper network, for example:

(1) Higher information about uncertain parameters (lesser deviation constant) leads to reduction in network cost,
(2) Conservative approach (higher value of $\Gamma$ ) can result in higher cost,
(3) There is cost saving in applying proposed model in comparison to classic network.

However, it is worth noting that the amount of cost saving is not generalizable and results are depends on network parameters.

## 6 Conclusions

In this paper, a robust reliable model to design relief logistics networks was proposed. The aim of the proposed model was to open the facilities (Shelters, UDCs and SDCs) in the pre-disaster phase and secondly, transportation of relief items and expected disruption cost in the post-disaster phase. In this study, in addition to the classical direct distribution, a practically-motivated distribution named as indirect distribution is considered to improve the fairness and correctness of distribution of relief items. As a result, for disrupted UDCs, a backup SDC was considered to support lost capacity of the UDC.

A robust decision making approach was considered to design relief network because on the one hand, the exact estimation and prediction of impact disaster on a network are tough and mangers face with information uncertainty in the pre-disaster phase of decision process, on the other hand, the enough relief items needed after a disaster such as earthquake are critical. Supply facilities play indispensable role to help and support affected people. However, these facilities may subject to disruption. Clearly, strengthening of supply facilities is an interesting strategy for managers. Therefore, to mitigate disruption impact, a budget-based strategy named as the soft-hardening is applied in this paper.

Developing an efficient solution approach to solve large scale network, considering procurement, multiple objectives and emergency medical services can be suggested as future research directions.

## References


Ahmadi, M., Seifi, A., & Tootooni, B. (2015). A humanitarian logistics model for disaster relief operation considering network failure and standard relief time: A case study on San Francisco district. *Transportation Research Part E: Logistics and Transportation Review*, *75*, 145–163.

Altay, N., & Green, W. G. (2006). OR/MS research in disaster operations management. *European Journal of Operational Research*, *175*, 475–493.

Azad, N., Saharidis, G. K. D., Davoudpour, H., Malekly, H., & Yektamaram, S. A. (2013). Strategies for protecting supply chain networks against facility and transportation disruptions: An improved Benders decomposition approach. *Annals of Operations Research*, *210*, 125–163.

Balcik, B., & Beamon, B. M. (2008). Facility location in humanitarian relief. *International Journal of Logistics Research and Applications*, *11*(2), 101–121.

Barzinpour, F., & Esmaeili, V. (2014). A multi-objective relief chain location distribution model for urban disaster management. *The International Journal of Advanced Manufacturing Technology*, *70*(5–8), 1291–1302.

Ben-Tal, A., & Nemirovski, A. (1999). Robust solutions of uncertain linear programs. *Operations Research Letters*, *25*(1), 1–13.

Bertsimas, D., & Sim, M. (2004). The price of robustness. *Operations Research*, *52*(1), 35–53.

Bozorgi-Amiri, A., Jabalameli, M. S., & Al-e-Hashem, S. M. J. M. (2013). A multi-objective robust stochastic programming model for disaster relief logistics under uncertainty. *OR Spectrum*, *35*(4), 905–933.

Bozorgi-Amiri, A., & Khorsi, M. (2016). A dynamic multi-objective location–routing model for relief logistic planning under uncertainty on demand, travel time, and cost parameters. *The International Journal of Advanced Manufacturing Technology*, *85*(5–8), 1633–1648.

Diatha, V. S. S. Y., & Sundar, K. (2015). Two substitutable perishable product disaster inventory systems. *Annals of Operations Research*, *233*, 517–534.

Döyen, A., Aras, N., & Barbarosoğlu, G. (2012). A two-echelon stochastic facility location model for humanitarian relief logistics. *Optimization Letters*, *6*(6), 1123–1145.

Duhamel, C., Santos, A. C., Brasil, D., Châtelet, E., & Birregah, B. (2016). Connecting a population dynamic model with a multi-period location-allocation problem for post-disaster relief operations. *Annals of Operations Research*, *247*(2), 693–713.

Hatefi, S. M., Jolai, F., Torabi, S. A., & Tavakkoli-Moghaddam, R. (2015). A credibility-constrained programming for reliable forward-reverse logistics network design under uncertainty and facility disruptions. *International Journal of Computer Integrated Manufacturing*, *28*(6), 664–678.

Hong, X., Lejeune, M. A., & Noyan, N. (2015). Stochastic network design for disaster preparedness. *IIE Transactions*, *47*(4), 329–357.

Huang, R., Kim, S., & Menezes, M. B. (2010). Facility location for large-scale emergencies. *Annals of Operations Research*, *181*(1), 271–286.

Jin, S., Jeong, S., Kim, J., & Kim, K. (2015). A logistics model for the transport of disaster victims with various injuries and survival probabilities. *Annals of Operations Research*, *230*(1), 17–33.

Kaur, H., & Singh, S. P. (2016). Sustainable procurement and logistics for disaster resilient supply chain. *Annals of Operations Research*, 1–46.

Kilci, F., Kara, B. Y., & Bozkaya, B. (2015). Locating temporary shelter areas after an earthquake: A case for Turkey. *European Journal of Operational Research*, *243*(1), 323–332.

Li, A. C. Y., Nozick, L., Xu, N., & Davidson, R. (2012). Shelter location and transportation planning under hurricane conditions. *Transportation Research Part E: Logistics and Transportation Review*, *48*(4), 715–729.

Mulvey, J. M., Vanderbei, R. J., & Zenios, S. A. (1995). Robust optimization of large-scale systems. *Operations research, 43*(2), 264–281.

Najafi, M., Eshghi, K., & Dullaert, W. (2013). A multi-objective robust optimization model for logistics planning in the earthquake response phase. *Transportation Research Part E: Logistics and Transportation Review*, *49*(1), 217–249.

Pan American Health Organization. (2001). *Humanitarian supply management and logistics in the health sector*.

Rath, S., & Gutjahr, W. J. (2014). A math-heuristic for the warehouse location-routing problem in disaster relief. *Computers and Operations Research*, *42*, 25–39.

Saadatseresht, M., Mansourian, A., & Taleai, M. (2009). Evacuation planning using multiobjective evolutionary optimization approach. *European Journal of Operational Research*, *198*(1), 305–314.

Salman, F. S., & Yücel, E. (2015). Emergency facility location under random network damage: Insights from the Istanbul case. *Computers and Operations Research*, *62*, 266–281.

Sharif, M. T., & Salari, M. (2015). A GRASP algorithm for a humanitarian relief transportation problem. *Engineering Applications of Artificial Intelligence*, *41*, 259–269.

Sheu, J. B., & Pan, C. (2014). A method for designing centralized emergency supply network to respond to large-scale natural disasters. *Transportation Research Part B: Methodological*, *67*, 320–343.

Tofighi, S., Torabi, S. A., & Mansouri, S. A. (2016). Humanitarian logistics network design under mixed uncertainty. *European Journal of Operational Research*, *250*(1), 239–250.

Trunick, P. A. (2005). Delivering relief to tsunami victims. *Logistics Today*, *46*(2), 1–3.

Yang, F., Yuan, Q., Du, S., & Liang, L. (2016). Reserving relief supplies for earthquake: a multi-attribute decision making of China Red Cross. *Annals of Operations Research*, *247*(2), 759–785.

Zhen, L., Wang, K., & Liu, H. (2015). Disaster relief facility network design in metropolises. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, *45*(5), 751–761.

Zokaee, S., Bozorgi-Amiri, A., & Sadjadi, S. J. (2016). A robust optimization model for humanitarian relief chain design under uncertainty. *Applied Mathematical Modelling*, *40*(17), 7996–8016.